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Co-Founders,
Confirmktkt

Make IRCTC great again!

Here's a startup that's battling one of everyone's worst nightmares – of booking a ticket successfully on IRCTC.

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Q Can you tell us how the idea to make this app came about?

The idea of starting Confirmktkt was conceptualised in 2014. Both Sripad and I were working in Bangalore at that time. Typically, we tried to go back home (Hyderabad) every weekend and train was the most convenient mode of transport. But, we hardly got our tickets confirmed due to the long waitlists. Just out of curiosity, we tried to dig deeper and noticed a particular pattern in the booking and cancellation trends of trains. We thought about applying this new-found knowledge to predict the waitlisted tickets. Once the initial experiments were successful, we decided to build the underlying software and launched

Confirmktkt the following year. Looking back, we can say that it was a process of trial and error.

Q What kind of data is used to train the ML algorithms? Is this training process continuous and ongoing?

We use the booking and cancellation information, type of train (local/passenger/long-distance), type of day (weekend/ weekday/holiday), source station, destination station and the route taken by the train to run our ML algorithms. However, the training process is continuous. Once the prediction is made for a PNR, we then log and cross-check it at the time when the train's final chart is prepared. If our prediction turns out to be inaccurate, then our algorithm learns from it and predicts better.

Q How does the partnership with IRCTC exactly work?

IRCTC has a B2C scheme for third parties to sell train tickets through their platforms. Under this partnership, we get access to the booking APIs which help us collect payments and book the tickets on our platform. This is pretty similar to how an individual user books his/her tickets on IRCTC. There's a fixed

commission (₹12+GST) on every ticket booked by third parties.

Q What was or is the toughest challenge that the app solved or is trying to solve?

Confirmktkt is addressing a unique problem for India. In India, getting a train ticket to travel can be a real nightmare. Billion-plus people try to get their hands on just 430 million tickets, thus resulting in a huge supply-demand gap. Through Confirmktkt's AI-driven prediction, we aim to provide a viable solution to the ticket booking woes of Indian train travellers.

Q What are some of the future plans for the app?

Confirmktkt was an information-only platform until March 2018, and then added the transactional feature to take the train ticket booking experience to the next level. We have also added multi-lingual support to our app to cater to those living outside urban settings. At present, we are working on developing voice-based services to capture the next set of internet users booking train tickets online.

Q What was the process that led to a 94% success rate?

We have accumulated around 7 years of historical data and made multiple changes in the algorithm that we used. Initially, we used to use very basic ML algorithms like regression, but later we moved to standard vector machines and then to neural networks.

Q How does Confirmktkt increase travel in trains?

In India, only 4% of the routes (4 Lakh trains) are directly connected. In order to increase the connectivity, we have built multimodal search which will build combinations of Train + Train, Train + Bus, Bus + Bus options where the direct connectivity is not available.

This has increased the bookable routes to 18 Lakhs. Currently, about 17% of the bookings happen via these combinations. This is unlike the past, when people were hesitant to book train tickets because of the waitlist. Thanks to the AI-based prediction, Confirmktkt has built confidence in the minds of passengers to book trains even on the waitlist. This has increased the waitlist tickets booking, thereby encouraging more people to travel via trains. **Q**